

I am writing to express my interest in the Research Assistant position focused on the characterisation of iPSCs neuronal models for Parkinson's disease. I hold a MSc in Biomedical Engineering, specialising in neurotechnology, from Imperial College London. My Master's project, which investigated the application of iPSC models for Alzheimer's disease, is highly relevant to this role.

In my project, we aimed to investigate the morphology and neural dynamics of cerebral organoids derived from both human and mouse cell lines during the early developmental stages. These 3D organoids were cultured and sectioned using in the lab. We employed various antibodies to identify neural progenitor cells, astrocytes, mature neurons, and the region-specific developments, which imaged with a confocal microscope. My primary focus was on conducting calcium imaging to track neural activities within a chamber equipped with a temperature regulation system. As working in a group project, I participated in and shadowed tissue culture, immunocytochemistry and calcium imaging procedures, which provided me with hands-on experience crucial for this position. I gained proficiency in wet lab techniques, microscopy operation techniques and image data analysis. Additionally, I developed skills in following and modifying standard procedures for cell culture experiments and maintaining neat records.

My academic background has given me a solid foundation in neuroscience, encompassing molecular mechanisms to higher level functions in the brain. This biological knowledge combined with mathematical methods in neural data analysis, computational modeling, and brain-machine interface, enhanced my understanding of the interconnections from cell biology to behavioural performances. Also, I gained strong programming skills for neural data analysis.

My previous laboratory experience spans various area, including electronics, physiological signals, and medical equipment calibration, which have refined my research skills. My previous experience as a research assistant honed my decision-making and communication skills, requiring self-motivation for project management and creativity for experimental design. Additionally, I led a group of seven peers in coursework for journal clubs during my Master's study, improving my teamwork, collaboration, and management skills. I believe these skills are transferable to this work.

I have a keen interest in understanding how brain functions and the development of neurological diseases. Studying the cell biology would be an excellent entry point for my future research in neuroscience. The research on up-to-date clinical practices using neuronal models of Parkinson's disease aligns well with my interests. The opportunity to join a dynamic and diverse group, collaborating with experts from various backgrounds would be invaluable for my personal development. I am excited about the prospect of contributing to your research team and advancing the diagnosis and treatment of Parkinson's disease.